Lectures 06 – 09

Chapter 4: Discrete-Time Markov Chains

References:

1. Oliver Ibe – Markov Processes for Stochastic Modeling (Chapter 3 and 4)
2. Sheldon Ross – Introduction to Probability Models (Chapter 4)

Objectives

1. Understand discrete-time Markov chains (DTMC) and its properties
2. Develop probability models associated with DTMC
3. Apply DTMC to understand Long-term behavior

Markov Processes

Applications

Markov processes are used to model various engineering systems:

1. Modeling wireless channels
2. Speech recognition, NLP, bioinformatics
3. Image segmentation, noise reduction and vision tasks
4. Foundation of reinforcement learning
5. Modeling Queueing systems
    a. Single/multi servers
    b. Single/multi queue
6. Network traffic modeling
7. Performance analysis of different computer and networking systems
    a. CSMA/CD or CSMA/CA
    b. TCP congestion control dynamics
8. Algorithm Analysis
    a. Cache replacement strategies (LRU)
    b. Probabilistic behavior of hashing, sorting and load balancing

Markov Process: Definition

A stochastic process is said to be a Markov Process, if it satisfies the Markov/Markovian properties

Markov Property:

Given the present state of a process, the future of the process is independent of the past.

The future of the process only depends on the present state

Present state of the process can be represented by

Only by the current state

By n states of the process including the current state [$(n-1)$ past states]

Accordingly, a Markov process is said to be an n-th order Markov Process

<u>First-order Markov process:</u>

If the future state of the Markov process is given only by its current state the process

It is said to be a first order Markov process

An SP $\{X(t), t \in T\}$ is a first-order Markov process, if for $0 < t_1 < t_2 < \cdots < t_n$,

the conditional CDF of $X(t_n)$ for given values of $X(t_0), X(t_1), \ldots, X(t_{n-1})$ depends on $X(t_{n-1})$

$$P[X(t_n) \leq x_n | X(t_{n-1}) = x_{n-1}, X(t_{n-2}) = x_{n-2}, \ldots, X(t_0) = x_0]$$
$$= P[X(t_n) \leq x_n | X(t_{n-1}) = x_{n-1}]$$

<u>Second-order Markov Process</u>: The future state of the process depends on both the current state and the immediate past state

<u>n-th order Markov Process</u>: Future state of the process depends on the current state and immediate past $(n-1)$ states.

<u>In this course, unless otherwise stated, we will consider first order Markov Processes only</u>

Like other SPs, Markov Processes are also classified according to the nature of the time parameter and the state space.

An SP with discrete states is called a chain, and a discrete state Markov process is called a Markov chain.

Markov processes can be either discrete-time or continuous-time.

Four basic types of Markov processes:

1. Discrete-time Markov chain (or discrete-time and discrete-state Markov process) – DTMC
2. Continuous-time Markov chain (or continuous-time and discrete-state Markov process) – CTMC
3. Discrete-time Markov process (or discrete-time and continuous-state Markov process) – DTMP
4. Continuous-time Markov process (or continuous-time and continuous-state Markov process) - CTMP

		State space	
		Discrete	Continuous
Time	Discrete	Discrete-time Markov chain	Discrete-time Markov process
	Continuous	Continuous-time Markov chain	Continuous-time Markov process

Recurrence Time and First Passage Time

- Recurrence Time: If T_i is the recurrence time of state i, then starting at state i, the process will enter state i for the first time in exactly n steps.

  $$\{T_{ii} = n\} = \{X_0 = i, X_1 \neq i, \dots, X_n = i\}$$

  We also define $f_{ii}(n)$ as the conditional probability that given the process is currently in state i, the first time it will enter into state i in exactly n transitions.

  $$f_{ii}(n) = P[X_n = i, X_{n-1} \neq i, \dots, X_2 \neq i, X_1 \neq i | X_0 = i]$$

- First Passage Time: The first passage time between state i and state j is the number of transitions n required for the process to enter state j for the first time, given that the process is currently in state i

$$\{T_{ij} = n\} = \{X_0 = i, X_1 \neq i, \dots, X_n = j\}$$

We also define $f_{ij}(n)$ as the conditional probability that given the process is currently in state i, the first passage time of the state j will be T_{ij},

that is, after exactly n transitions the process will enter state j for the first time, given that it is currently in state i.

$$f_{ij}(n) = p[X_n = j, X_{n-1} \neq j, \dots, X_2 \neq j, X_1 \neq j | X_0 = i]$$

Applications of Markov Chains/Process:

Discrete-time Markov Chains:

Branching Processes:

- System starts with a finite set of elements
- Each element
  - Disappears with probability p_0 or
  - Produce k other similar elements with probability $p_k, k = 1,2, \ldots$
  - Each of these k elements behaves like their parents
- X_n denotes the population size after n events
- X_n denotes the population size after n events
- MC $\{X_n, n \geq 0\}$ is a branching process

Markov Decision Process (MDP):

- A decision maker known as the agent takes a sequence of decisions with uncertain outcomes.
- Each decision results in a reward or a penalty
- Goal is to find an optimal sequence of decisions (actions) that maximizes the reward for a given time interval.

<u>Continuous-time Markov Chains</u>:

Queueing Systems:

- A queue is a waiting line
- It consists of one or more servers that server the customers
- Customers arrive following a well-defined SP
- If an arriving customer finds an idle server, it is served immediately, otherwise the customer waits in a queue
- Let X_n denote the number of customers in the queue at time period n
- If the arrival process is Poisson, and the service times are exponentially distributed, the $\{X(t), t \geq 0\}$ is a CTMC.
- Services may be provided in multiple stages – outcome of one stage is sent to the input of the next – network of queues.

CT Markov Decision Process:

In a discrete-time MDP, actions are taken at discrete time intervals, whereas in a continuous-time MDP, actions are taken over exponentially distributed time intervals.

Continuous-state Markov Process:

Brownian Motion Process

- A continuous-time continuous-state process is the Brownian Motion process, first proposed by the botanist Robert Brown, who observed that the pollen particles suspended in a fluid moved in an irregular random manner.
- Einstein obtained the equation for Brownian Motion, and Wiener established the mathematical foundation for Brownian motion as a stochastic process.
- Has been successfully used to describe thermal noises in electric circuitry.

Diffusion Process

- A related process of the Brownian process is the diffusion process
- Particles are transported form one part of a system to another as result of random Motion.
- The direction of the motion of particles is from a region of higher concentration to a region of lower concentration of the particles.
- Brownian motion is the random motion of molecules, and the direction of motion of these molecules is random.
- Diffusion is the movement of particles from areas of high concentration to areas of low concentration.
- Brownian motion is completely random while diffusion does not occur in a homogeneous medium where there is no concentration gradient.
- Brownian motion may be considered a probabilistic model of diffusion in a homogenous medium.

Discrete-time Markov Chains (DTMC)

Introduction

Let $k = \{0,1,2, \dots\}$ be a discrete set of time, and

$S = \{s_1, s_2, \dots, m\}$ be a discrete set of states

An MP starts at $k = 0$ at one of these states

At each time period, the process moves from one state to another, called a *transition* or a *step*

X_k be the state of the process at time period k is a RV,

Representing its state at time period k, which is one of S

Therefore, a probability exists representing, the likelihood of moving to a state j, from the current state i

The process might stay in the same state at the next time period

Such a transition from state i to state i is called self transition

One-step Transition Probability

The one-step transition probability of a chain from state i to state j is

$$p_{ij} = P[X_k = j | X_{k-1} = i, X_{k-2} = n, \dots, X_0 = m] \qquad i, j \in S, k \geq 0 \text{ , or}$$

$$= P[X_k = j | X_{k-1} = i]$$

$$p_{ijk} = P[X_k = j | X_{k-1} = i, X_{k-2} = n, \dots, X_0 = m] \qquad i, j \in S, k \geq 0$$

$$= P[X_k = j | X_{k-1} = i]$$

Markov Property:

Future state of the chain depends on the past states

First-order: Future state depends on the current state of the chain

2nd-order: Future state depends on the current and the immediate past states

Non-homogeneous MC:

$$P[X_k = j | X_{k-1} = i, X_{k-2} = n, \ldots, X_0 = m] = P[X_k = j | X_{k-1} = i] = p_{ijk}$$

Homogeneous MC:

$$P[X_k = j | X_{k-1} = i, X_{k-2} = n, \ldots, X_0 = m] = P[X_k = j | X_{k-1} = i] = p_{ij}$$

Homogeneous state-transition probability p_{ij} satisfies:

1. $0 \leq p_{ij} \leq 1$
2. $\sum_j p_{ij} = 1, \quad i = 1, 2, \ldots$

$$
\begin{aligned}
&P[X_k = j, X_{k-1} = i_1, X_{k-2} = i_2, \ldots, X_0 = i_k] \\
&\quad = P[X_k = j | X_{k-1} = i_1, X_{k-2} = i_2, \ldots, X_0 = i_k] P[X_{k-1} = i_1, X_{k-2} = i_2, \ldots, X_0 = i_k] \\
&\quad = P[X_k = j | X_{k-1} = i_1] P[X_{k-1} = i_1, X_{k-2} = i_2, \ldots, X_0 = i_k] \\
&\quad = P[X_k = j | X_{k-1} = i_1] P[X_{k-1} = i_1 | X_{k-2} = i_2, \ldots, X_0 = i_k] P[X_{k-2} = i_2, \ldots, X_0 = i_k] \\
&\quad = P[X_k = j | X_{k-1} = i_1] P[X_{k-1} = i_1 | X_{k-2} = i_2] P[X_{k-2} = i_2, \ldots, X_0 = i_0] \\
&\quad = P[X_k = j | X_{k-1} = i_1] P[X_{k-1} = i_1 | X_{k-2} = i_2] \cdots P[X_1 = i_{k-1} | X_0 = i_k] P[X_0 = i_k] \\
&\quad = p_{i_1 j} p_{i_2 i_1} p_{i_3 i_2} \ldots p_{i_k i_{k-1}} P[X_0 = i_k]
\end{aligned}
$$

Specifying a MC:

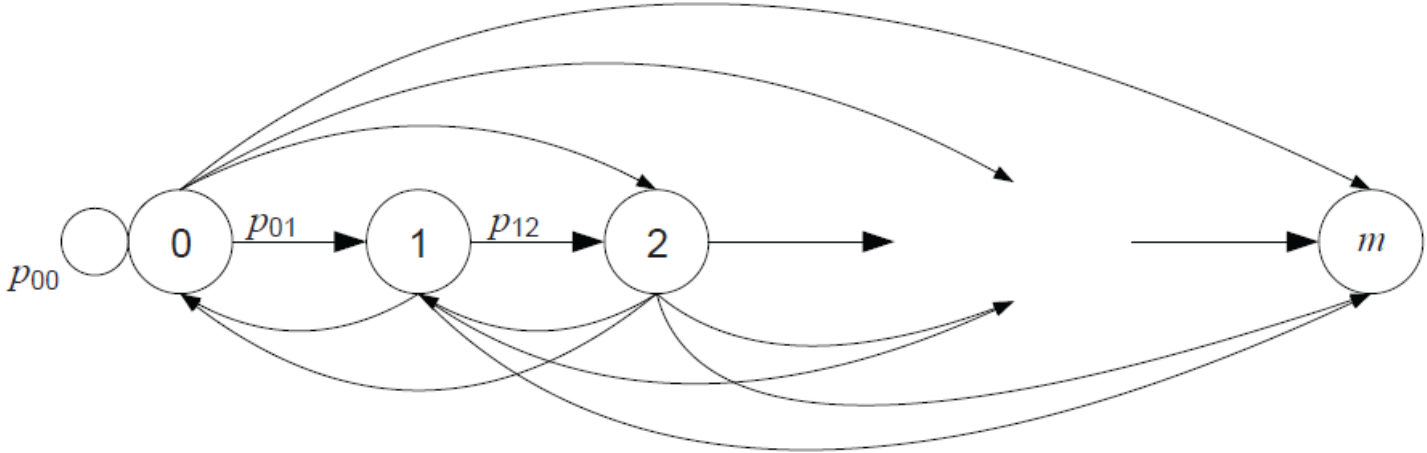
Probability model of a Markov chain is specified by

1. State space
2. State-transition probabilities

State transition probability matrix:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$$

State-transition graph/diagram



DTMC: A discrete time stochastic process that satisfies the Markov Property

Markov Property: The Future of the process depends only on the present

One-step transition probability (p_{ij}): Conditional probability that the process will be in state j at time t given that the process currently is in state i

Transition probability matrix (P)

Transition probability graph

Example Markov Chains

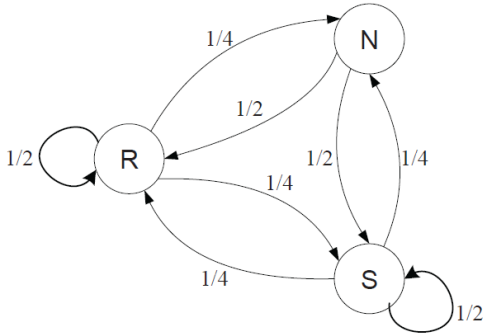
Example 3.1 (Weather in the land of Oz)**.** The land of Oz is blessed by many things, but not by good weather. They never have two nice days in a row. If they have one nice day, they are just likely to have snow day as a rainy day in the next day. If they have a snowy day or a rainy day, they have an even chance of having the same in the next day. If there is a change from snow or rain, only half of the time is this a change to a nice day. Model the system as a discrete time Markov chain with appropriate states and show the transition probability matrix and transition probability graph of the chain.

state 0 - a rainy day (R)

state 1 - a nice day (N)

state 2 - a snowy day (S)

$$\mathbf{P} = \begin{array}{c|ccc|} & R & N & S \\ R & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ N & \frac{1}{2} & 0 & \frac{1}{2} \\ S & \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{array}$$

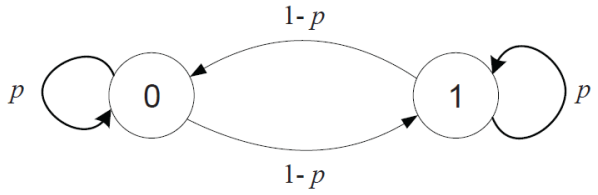

N
R
S
1/4
1/2
1/2
1/2
1/4
1/4
1/4
1/2

Example 3.2 (A communication System)**.** Consider a communication system which transmits the digit 0 and 1. Each digit transmitted must pass through several stages, at each of which there is a probability p that the digit entered will be unchanged when it leaves. Model the system as a discrete time Markov chain with appropriate states and show the transition probability matrix and transition probability graph of the chain.

State 0 -> current digit is 0

State 1 -> current digit is 1

$$P = \begin{vmatrix} p & 1-p \\ 1-p & p \end{vmatrix}$$



Example 3.3 (Transforming a process into a Markov chain)**.** Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days then it will rain tomorrow with probability 0.2.

a. Why the system is not a Markov chain?

b. Show how the system can be transformed into a Markov chain.

c. Show the transition matrix P for the transformed Markov chain.

d. Draw the transition probability graph for the transformed Markov chain.

state 0 - raining (R)

state 1 - not raining (N)

The state of the chain is given by

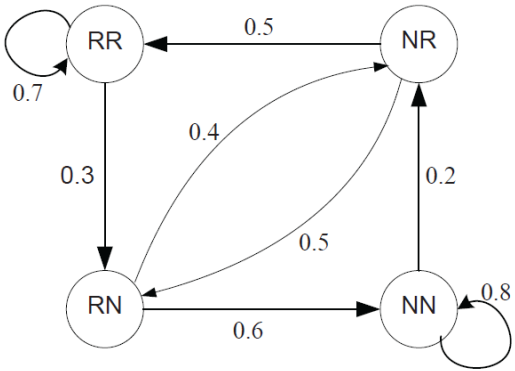
state 0 - if it rained both today and yesterday (RR)

state 1 - if it rained today but not yesterday (NR)

state 2 - if it rained yesterday but not today (RN)

state 3 - if it did not rain both today and yesterday (NN

$$P = \begin{array}{c} RR \\ NR \\ RN \\ NN \end{array} \begin{vmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{vmatrix}$$



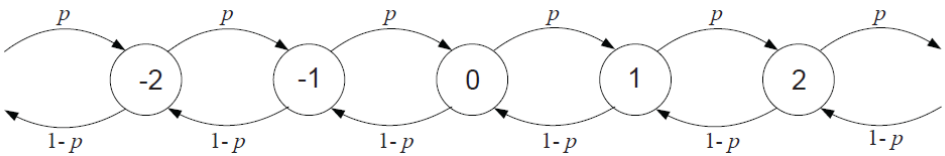
Example 3.4 (Random Walk)**.** Consider an individual is walking on a straight line. Assume that the individual, at each point of time, either takes one step to the right with probability p, or one step to the left with probability $1 - p$. The movement of the individual is known as random walk. Model the system as a Markov chain.

$$i = 0, \pm 1, \pm 2, \pm 3, \cdots.$$

$$p_{i,i+1} = p$$

$$p_{i,i-1} = 1 - p$$

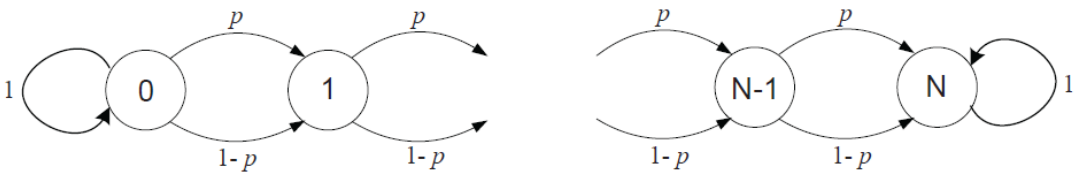
$$p_{i,i+1} = p = 1 - p_{i,i-1}$$



Example 3.5 (A Gambling Model)**.** Consider a gambler who, at each play of the game, either wins \$1 with probability p, or losses \$1 with probability $1-p$. Furthermore, assume that the gambler quits playing the game when he goes broke or when he attains a fortune of \$N, then the the gambler's fortune is a Markov chain with transition probabilities

$$p_{i,i+1} = p = 1 - p_{i,i-1} \qquad i = 1, 2, \cdots, N$$

$$p_{0,0} = p_{N,N} = 1$$



n-step transition probability

Conditional prob. that the process will be in state j, after exactly n transitions, given that it is presently in state i, $p_{ij}(n)$

$$p_{ij}(n) = P[X_{m+n} = j | X_m = i]$$

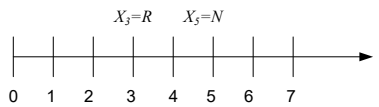
$$p_{ij}(0) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

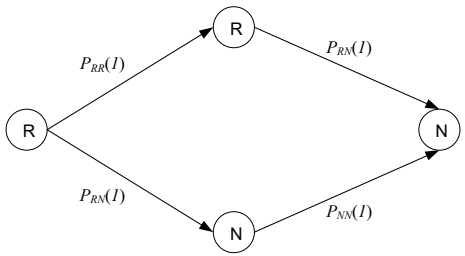
$$p_{ij}(1) = p_{ij}$$

Consider 2-step transition probability

$$p_{ij}(2) = P[X_{m+2} = j | X_m = i]$$

$$\begin{aligned} p_{ij}(2) &= P[X_2 = j | X_0 = i] = \sum_k P[X_2 = j, X_1 = k | X_0 = i] \\ &= \sum_k P[X_2 = j | X_1 = k, X_0 = i] P[X_1 = k | X_0 = i] \\ &= \sum_k P[X_2 = j | X_1 = k] P[X_1 = k | X_0 = i] = \sum_k p_{kj} p_{ik} \\ &= \sum_k p_{ik} p_{kj} \end{aligned}$$





2-step transition probabilities

$$p_{RN}(2) = p_{RR}p_{RN} + p_{RN}p_{NN}$$

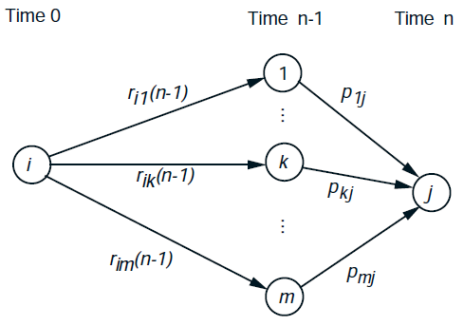
$$p_{RR}(2) = p_{RR}p_{RR} + p_{RN}p_{NR}$$

$$p_{ij}(2) = \sum_{k=1}^{2} p_{ik}p_{kj}$$

$$P^2 = P \cdot P$$

n-step transition probabilities

Sum of probabilities of all paths that start at state i and end in state j in exactly n transitions



Chapman-Kolmogorov Equations: For all $0 < r < n$

$$p_{ij}(n) = \sum_k p_{ik}(r) p_{kj}(n-r)$$

$$\begin{aligned}
p_{ij}(n) &= P[X_n = j | X_0 = i] = \textstyle\sum_k P[X_n = j, X_r = k | X_0 = i] \\
&= \textstyle\sum_k P[X_n = j | X_r = k, X_0 = i] P[X_r = k | X_0 = i] \\
&= \textstyle\sum_k P[X_n = j | X_r = k] P[X_r = k | X_0 = i] = \textstyle\sum_k p_{kj}(n-r) p_{ik}(r) \\
&= \textstyle\sum_k p_{ik}(r) p_{kj}(n-r)
\end{aligned}$$

$p_{ij}(n)$ is the ij-th entry in the matrix P^n

$$P^n = \begin{bmatrix} p_{11}(n) & p_{12}(n) & \dots & p_{1N}(n) \\ p_{21}(n) & p_{22}(n) & \dots & p_{2N}(n) \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1}(n) & p_{N2}(n) & \dots & p_{NN}(n) \end{bmatrix}$$

Properties of states

Accessibility: state j is accessible from state i, if

Starting from state i, the process can ever enter state j

$p_{ij}(n) > 0$, for some n

$i \rightarrow j$ denote accessibility of j from i

Communication: two states i and j communicate if

States i and j are accessible from the other

$i \rightarrow j$ and $j \rightarrow i$,

$i \leftrightarrow j$ denotes communications

Class: Communication divides the states into different classes

Communication is a class property

Two states that communicate are in the same class

All states in a single class communicate with each other

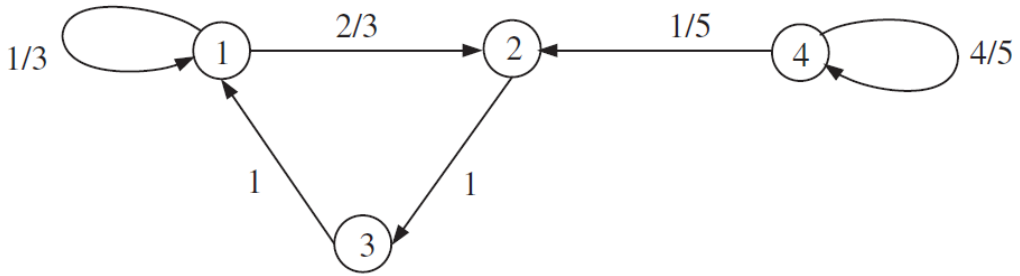
Closed communicating class – not accessible from outside the class

Irreducible Markov Chains: all states communicate – only one class

Classifications of states:

<u>Transient state</u>: positive probability exists that the process will not return to the state again, after it leaves the state

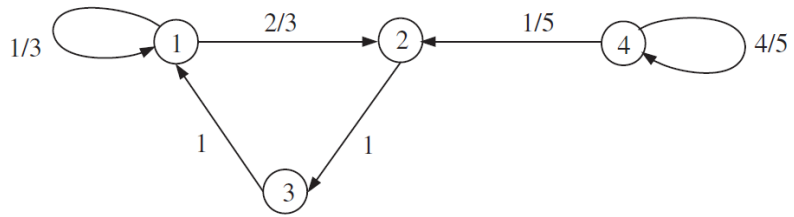
Consider the Markov chain with the state-transition diagram shown in Figure 4.2. State 4 is the only transient state; states 1, 2, and 3 are recurrent states; and there is no periodic state. There is a single chain, which is $\{1, 2, 3\}$.



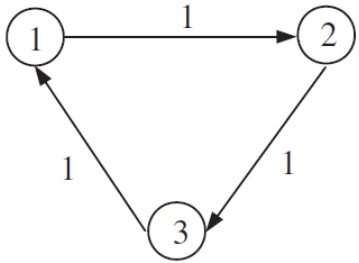
<u>Recurrent states</u>: with a probability 1, the process will return to the state again, after it leaves the state

- Periodic: if there exists an integer $d > 1$, such that $p_{ij}(n) = 0$, if n is not multiple of d
- If $d = 1$, then a recurrent state is aperiodic
- A recurrent state is positive recurrent, if the *expected recurrence* time is finite, otherwise it is null recurrent

Consider the Markov chain with the state-transition diagram shown in Figure 4.2. State 4 is the only transient state; states 1, 2, and 3 are recurrent states; and there is no periodic state. There is a single chain, which is $\{1, 2, 3\}$.



The Markov chain whose state-transition diagram is shown in Figure 4.4 has a single chain $\{1, 2, 3\}$, and the three states are periodic with a period of 3.



Ergodic state: Positive recurrent and aperiodic states are called ergodic states

Ergodic MC: A MC chain consisting of ergodic states is called an ergodic chain

<u>First Passage Time</u>: The first passage time of state j from state i is a RV

Starting from state i, the process visits state j for the first time in n transitions

$$T_{ij} = \min(n \geq 1 : X_n = j | X_0 = i)$$

The <u>probability of first passage</u> from state i to state j in n steps/transitions

$$f_{ij}(n) = \sum_{l \neq j} p_{il} f_{lj}(n-1)$$

The probability of first passage from state i to state j is a conditional probability that the process will ever enter j given that it is initially in state i

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n)$$

The *recurrence time* T_i of state i is the random variable that is defined as follows:

$$T_i = \min \{n \geq 1\colon\ X_n = i \text{ given that } X_0 = i\}$$

That is, T_i is the time until the process returns to state i, given that it was initially in state i. The probability that the first return to state i occurs after n transitions, $f_{ii}(n)$, is defined by

$$f_{ii}(n) = P[T_i = n]$$

A state i is said to be *transient* if, given that we start in state i, there is a nonzero probability that we will never return to i. For a transient state i,

$$P[T_i < \infty] = \sum_{n=1}^{\infty} f_{ii}(n) = f_{ii} < 1$$

A state i in which $f_{ii} = P[T_i < \infty] = 1$ is called a *recurrent* state. The *mean recurrence time* of state i, m_i, is given by

$$m_i = E[T_i] = \sum_{n=1}^{\infty} n f_{ii}(n)$$

A state i is called *absorbing* or *trapping* state if it is impossible to leave this state once the process enters the state; that is, when the process enters an absorbing state, it is "trapped" and cannot leave the state. Therefore, the state i is absorbing if and only if

$$p_{ii} = 1, \quad \text{and} \quad p_{ij} = 0 \quad \text{for} \quad j \neq i$$

Example 4.1

Consider the Markov chain with the state-transition diagram shown in Figure 4.2. State 4 is the only transient state; states 1, 2, and 3 are recurrent states; and there is no periodic state. There is a single chain, which is $\{1, 2, 3\}$.

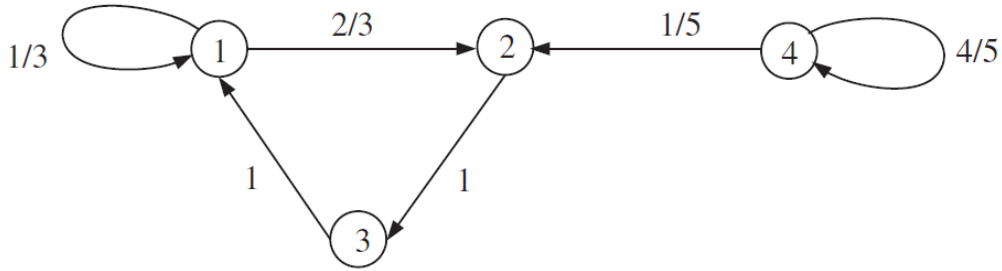


Figure 4.2 State-transition diagram for Example 4.1.

Example 4.2

Consider the state-transition diagram of Figure 4.3. The transition is now from state 2 to state 4 instead of from state 4 to state 2. For this case, states 1, 2, and 3 are now transient states because when the process enters state 2 and makes a transition to state 4, it does not return to these states again. Also, state 4 is a trapping (or absorbing) state because when the process enters the state, the process never leaves the state. As stated in the definition, we identify a trapping state from the fact that, as in this example, $p_{44} = 1$ and $p_{4k} = 0$ for k not equal to 4.

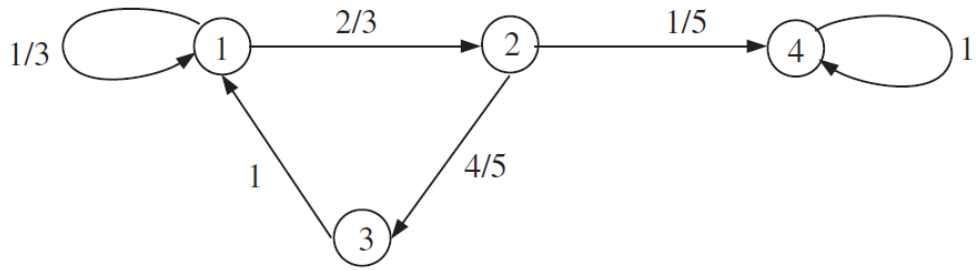


Figure 4.3 State-transition diagram for Example 4.2.

Example 4.3

The Markov chain whose state-transition diagram is shown in Figure 4.4 has a single chain $\{1, 2, 3\}$, and the three states are periodic with a period of 3.

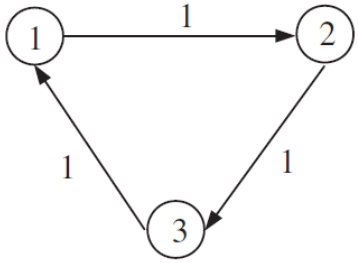
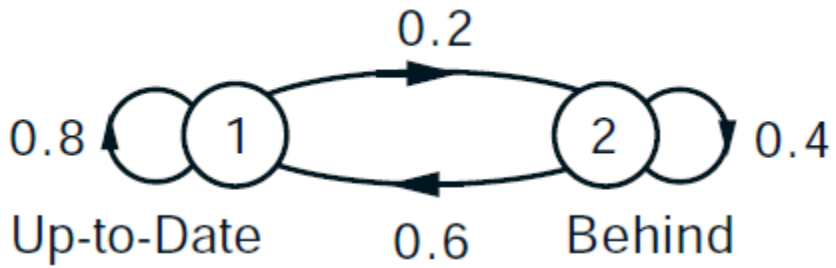


Figure 4.4 State-transition diagram for Example 4.3.

Limiting state probabilities



	UpD	B
UpD	0.8	0.2
B	0.6	0.4

$r_{ij}(1)$

.76	.24
.72	.28

$r_{ij}(2)$

.752	.248
.744	.256

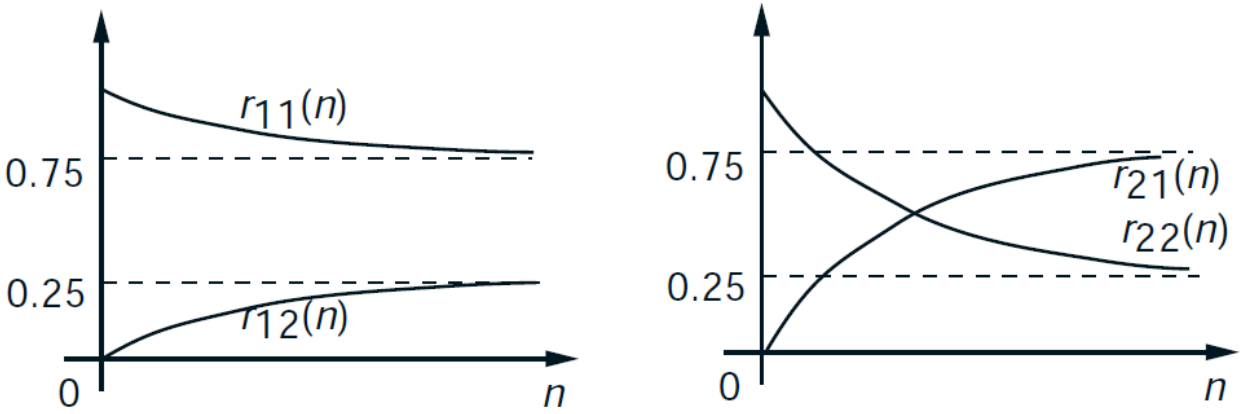
$r_{ij}(3)$

.7504	.2496
.7488	.2512

$r_{ij}(4)$

.7501	.2499
.7498	.2502

$r_{ij}(5)$



n-step transition probabilities as a function of the number n of transitions

If we define $P[X(0) = i]$ as the probability that the process is in state i before it makes the first transition, then the set $\{P[X(0) = i]\}$ defines the initial condition for the process, and for an N-state process,

$$\sum_{i=1}^{N} P[X(0) = i] = 1$$

Let $P[X(n) = j]$ denote the probability that it is in state j at the end of the first n transitions, then for the N-state process,

$$P[X(n) = j] = \sum_{i=1}^{N} P[X(0) = i] p_{ij}(n)$$

Steady-State Convergence Theorem

Consider a Markov chain with a single recurrent class, which is aperiodic. Then, the states j are associated with steady-state probabilities π_j that have the following properties.

(a)
$$\lim_{n\to\infty} r_{ij}(n) = \pi_j, \qquad \text{for all } i,\ j.$$

(b) The π_j are the unique solution of the system of equations below:

$$\pi_j = \sum_{k=1}^{m} \pi_k p_{kj}, \qquad j = 1, \ldots, m,$$
$$1 = \sum_{k=1}^{m} \pi_k.$$

(c) We have

$$\pi_j = 0, \qquad \text{for all transient states } j,$$
$$\pi_j > 0, \qquad \text{for all recurrent states } j.$$

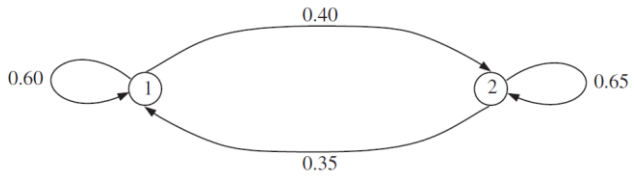


Figure 4.5 State-transition diagram for Example 4.4.

$$\pi_1 = 0.6\pi_1 + 0.35\pi_2$$
$$\pi_2 = 0.4\pi_1 + 0.65\pi_2$$
$$1 = \pi_1 + \pi_2$$

$$\pi_1 = 0.6\pi_1 + 0.35\pi_2$$

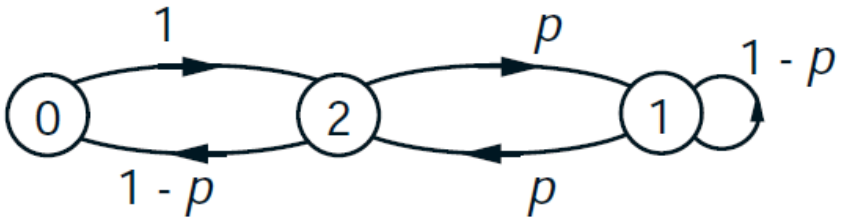
$$1 = \pi_1 + \pi_2$$

$$\pi_1 = \frac{0.35}{0.4}\pi_2 = \frac{7}{8}\pi_2$$

$$1 = \frac{7}{8}\pi_2 + \pi_2 = \frac{15}{8}\pi_2 => \pi_2 = \frac{8}{15}$$

$$\pi_1 = \frac{7}{15}$$

Example 6.5. An absent-minded professor has two umbrellas that she uses when commuting from home to office and back. If it rains and an umbrella is available in her location, she takes it. If it is not raining, she always forgets to take an umbrella. Suppose that it rains with probability p each time she commutes, independently of other times. What is the steady-state probability that she gets wet on a given day?



$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1-p & p \\ 1-p & p & 0 \end{bmatrix}$$

$$\pi_0 = (1-p)\pi_2, \quad \pi_1 = (1-p)\pi_1 + p\pi_2, \quad \pi_2 = \pi_0 + p\pi_1.$$

$$\pi_0 = \frac{1-p}{3-p}, \quad \pi_1 = \frac{1}{3-p}, \quad \pi_2 = \frac{1}{3-p}.$$

Stochastic and Doubly Stochastic Method

Probability transition matrices are stochastic matices because for any row

$$\sum_j p_{ij} = 1$$

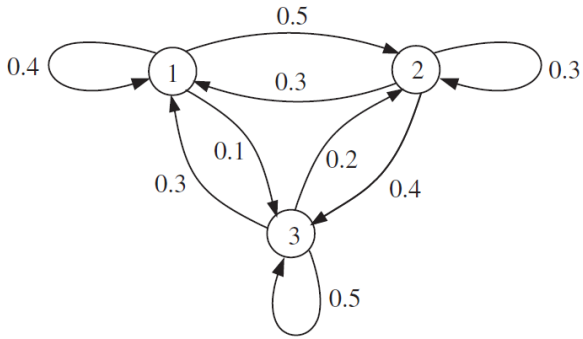
Doubly stochastic matrix, columns also sum to 1,

$$\sum_i p_{ij} = 1$$

Theorem 4.1 If P is a doubly stochastic matrix associated with the transition probabilities of a Markov chain with N states, then the limiting-state probabilities are given by $\pi_i = 1/N, \quad i = 1, 2, \ldots, N$.

Example 4.5

Find the transition probability matrix and the limiting-state probabilities of the process represented by the state-transition diagram shown in Figure 4.6.



Solution

The transition probability matrix is given by

$$P = \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.3 & 0.4 \\ 0.3 & 0.2 & 0.5 \end{bmatrix}$$

It can be seen that each row of the matrix sums to 1, and each column also sums to 1; that is, it is a doubly stochastic matrix. Because the process is an irreducible, aperiodic Markov chain, the limiting-state probabilities exist and are given by $\pi_1 = \pi_2 = \pi_3 = 1/3$.

Sojourn Time

States with $p_{ii} > 0$, process might remain in the same state for more than one time period.

What is the probability that the process will in such states for exactly d time period?

D_i denote the number of time periods that the process remains in state i before making a transition, then the PMF of D_i is

$$p_{D_i}(d) = P[D_i = d] = P[X_0 = i, X_1 = i, X_2 = i, \ldots, X_{d-1} = i, X_d \neq i]$$
$$= p_{ii}^{d-1}(1 - p_{ii})$$

$$D_i \sim geom(1 - p_{ii})$$

Expected sojourn time is $\quad E[D_i] = \frac{1}{1-p_{ii}}$

Steady-State Probabilities as Expected State Frequencies

For a Markov chain with a single class that is aperiodic, the steady-state probabilities π_j satisfy

$$\pi_j = \lim_{n \to \infty} \frac{v_{ij}(n)}{n},$$

where $v_{ij}(n)$ is the expected value of the number of visits to state j within the first n transitions, starting from state i.

π_j is the long-term expected fraction of time state of the chain is j

$\pi_j p_{jk}$ is the long-time expected fraction of transitions from j to k

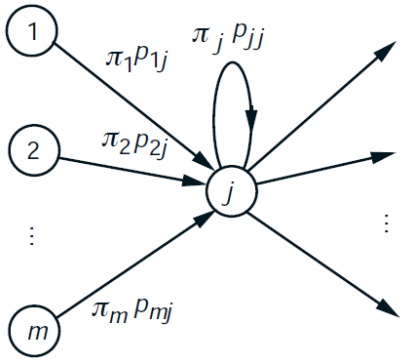
Expected Frequency of a Particular Transition

Consider n transitions of a Markov chain with a single class that is aperiodic, starting from a given initial state. Let $q_{jk}(n)$ be the expected number of such transitions that take the state from j to k. Then, regardless of the initial state, we have

$$\lim_{n\to\infty} \frac{q_{jk}(n)}{n} = \pi_j p_{jk}.$$

Expected frequency π_j is equal to the sum of expected frequencies $\pi_k p_{kj}$

$$\pi_j = \sum_{k=1}^{m} \pi_k p_{kj}$$



<u>Absorbing Markov Chains:</u> If there is at least one absorbing state in the MC

The most obvious question that can be asked about an absorbing Markov chain are:

1. What is the probability that hte process will eventually reach an absorbing state?
2. What is the probability that the process will end up in a given absorbing state?
3. On the average, how long will it take for the process to be absorbed?
4. On the average, how many times will the process be in each transient state?

The transition matrix of the markov chain is given by

$$\mathbf{P} = \left|\begin{array}{cccc|ccc} p_{11} & p_{12} & \cdots & p_{1t} & p_{1,t+1} & \cdots & p_{1,t+r} \\ p_{11} & p_{12} & \cdots & p_{1t} & p_{1,t+1} & \cdots & p_{1,t+r} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ p_{11} & p_{12} & \cdots & p_{1t} & p_{1,t+1} & \cdots & p_{1,t+r} \\ \hline 0 & 0 & \cdots & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 1 \end{array}\right|$$

The transition probability matrix $\mathbf{P}$ have the following canonical form:

$$\mathbf{P} = \left|\begin{array}{c|c} \mathbf{Q} & \mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array}\right|$$

where, $\mathbf{Q}$ is an t-by-t matrix, $\mathbf{R}$ is an t-by-r matrix, $\mathbf{I}$ is an r-by-r identity matrix, and $\mathbf{0}$ is an t-by-t zero matrix.

The n-step transition probability matrix $\mathbf{P}^n$ of an absorbing Markov chain has the following form:

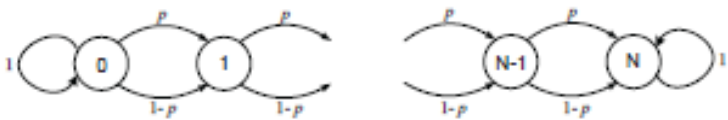
$$\mathbf{P} = \left|\begin{array}{c|c} \mathbf{Q}^n & * \\ \hline \mathbf{0} & \mathbf{I} \end{array}\right|$$

For an absorbing Markov chain, the inverse of the matrix $\mathbf{I} - \mathbf{Q}$ always exists. Let N denote the inverse matrix of $\mathbf{I} - \mathbf{Q}$, i.e., $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$. The matrix $\mathbf{N}$ is called the *fundamental matrix* $\mathbf{P}$.

Example 3.9. Consider the Gambling model in Example 3.5, and assume $p = 0.5$ and $n = 4$. Find the fundamental metrix for the absorbing Markov chain.

Solution. The transition probability matrix is given by

$$\mathbf{P} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \left\| \begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{array} \right\| \end{array}$$



$$
\mathbf{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 0 \\ 4 \end{array}
\begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 0 & 4 \end{array} \\ \left\| \begin{array}{ccc|cc} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \hline 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right\| \end{array}
$$

$$\mathbf{Q} = \begin{Vmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{Vmatrix}$$

$$\mathbf{I} - \mathbf{Q} = \begin{Vmatrix} 1 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 1 \end{Vmatrix}$$

Now, the inverse of $\mathbf{I} - \mathbf{Q}$, i.e., $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$, is

$$\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{Vmatrix} \frac{3}{2} & 1 & \frac{1}{2} \\ 1 & 2 & 1 \\ \frac{1}{2} & 1 & \frac{3}{2} \end{Vmatrix} \end{array}$$

Hence, the time to absorption of the transient states of the chain is

$$\mathbf{t} = \mathbf{N}\mathbf{c} = \begin{Vmatrix} \frac{3}{2} & 1 & \frac{1}{2} \\ \frac{3}{2} & 1 & \frac{1}{2} \\ \frac{3}{2} & 1 & \frac{1}{2} \end{Vmatrix} \cdot \begin{Vmatrix} 1 \\ 1 \\ 1 \end{Vmatrix}$$

$$= \begin{Vmatrix} 3 \\ 4 \\ 3 \end{Vmatrix}$$

Hence, the absorption probability of chain in state j starting from state i is given by

$$\mathbf{B} = \mathbf{NR} = \begin{Vmatrix} \frac{3}{2} & 1 & \frac{1}{2} \\ \frac{3}{2} & 1 & \frac{1}{2} \\ \frac{3}{2} & 1 & \frac{1}{2} \end{Vmatrix} \cdot \begin{Vmatrix} \frac{1}{2} & 0 \\ 0 & 0 \\ 0 & \frac{1}{2} \end{Vmatrix}$$

$$= \begin{Vmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{Vmatrix}$$

First Passage and Recurrence Times

$$T_{ij} = \min\{n \geq 1: \ X_n = j | X_0 = i\}$$

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n)$$

$$f_{ij}(n) = \sum_{l \neq j} p_{il} f_{lj}(n-1)$$

$i = j$, first passage time becomes recurrence time

Mean first passage time, m_{ij}

$$m_{ij} = \sum_{n=1}^{\infty} n f_{ij}(n) = 1 + \sum_{k \neq j} p_{ik} m_{kj} \quad i \neq j$$

Mean first passage time from i to j is the holding time in state i plus mean first passage time for j

$$m_{ii} = \sum_{n=1}^{\infty} n f_{ii}(n) = 1 + \sum_{k \neq i} p_{ik} m_{ki}$$

Example 4.7

$$P = \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.3 & 0.4 \\ 0.3 & 0.2 & 0.5 \end{bmatrix}$$

$$m_{13} = 1 + p_{11}m_{13} + p_{12}m_{23}.$$

$$\begin{aligned} m_{13} &= 1 + p_{11}m_{13} + p_{12}m_{23} \\ &= 1 + 0.4m_{13} + 0.5m_{23} \Rightarrow 0.6m_{13} = 1 + 0.5m_{23} \\ m_{23} &= 1 + p_{21}m_{13} + p_{22}m_{23} \\ &= 1 + 0.3m_{13} + 0.2m_{23} \Rightarrow 0.8m_{23} = 1 + 0.3m_{13} \end{aligned}$$

$$m_{13} = 3.939$$

$$m_{23} = 2.737$$

Recurrence time of state 1 m_{11}

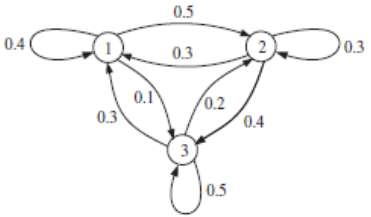
$$m_{11} = 1 + p_{12}m_{21} + p_{13}m_{31} = 1 + 0.5m_{21} + 0.1m_{31}$$

$$m_{21} = 1 + p_{22}m_{21} + p_{23}m_{31}$$

$$= 1 + 0.3m_{21} + 0.4m_{31} \Rightarrow 0.7m_{21} = 1 + 0.4m_{31}$$

$$m_{31} = 1 + p_{32}m_{21} + p_{33}m_{31}$$

$$= 1 + 0.2m_{21} + 0.5m_{31} \Rightarrow 0.5m_{31} = 1 + 0.2m_{21}$$



The solution to the system of equations is

$$m_{11} = 1.7187$$

$$m_{21} = 2.8125$$

$$m_{31} = 3.1250$$